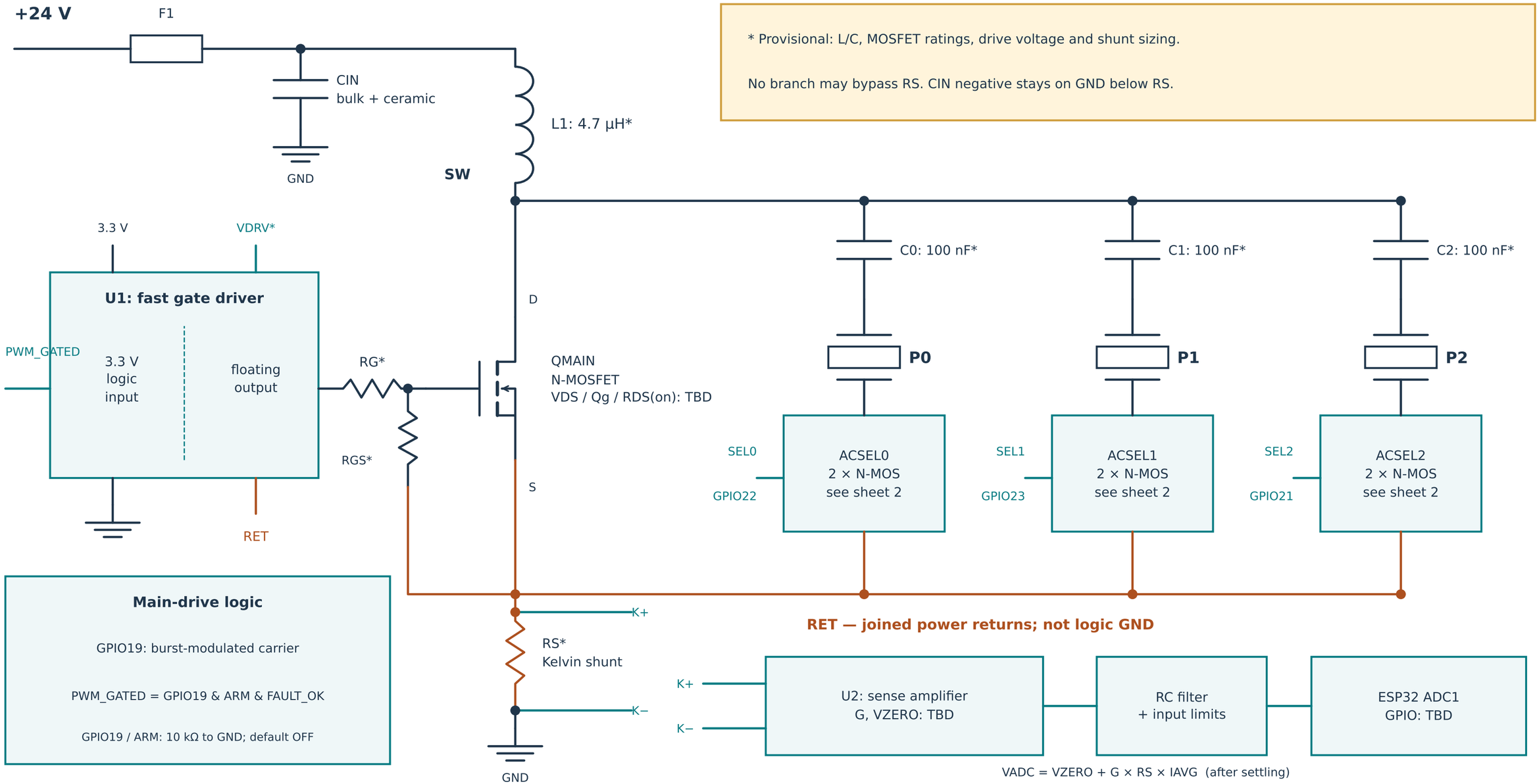
ESP32 / 24 V / three-piezo atomizer

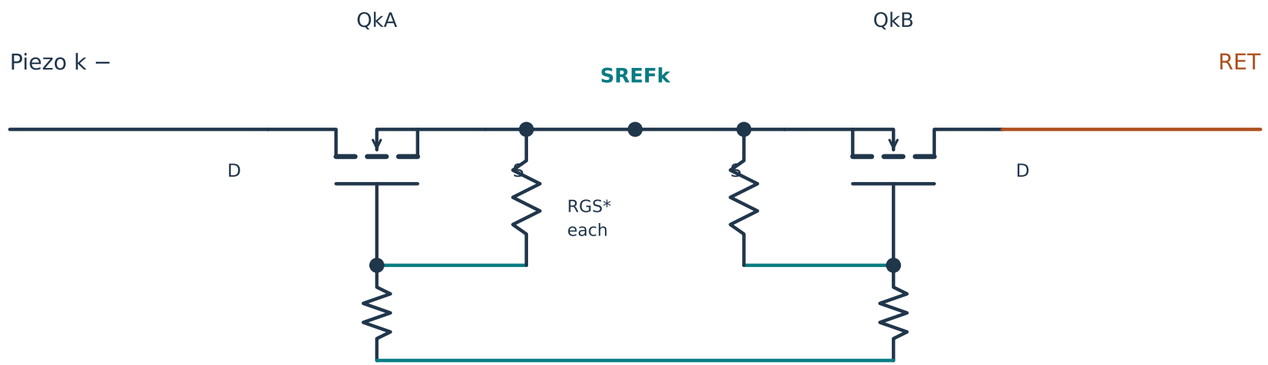
Sheet 1 — shared drive stage, independent AC selectors, and average input-current feedback



Selector drive and current-feedback details

Sheet 2 — repeat the selector cell three times; component choices remain to be finalized

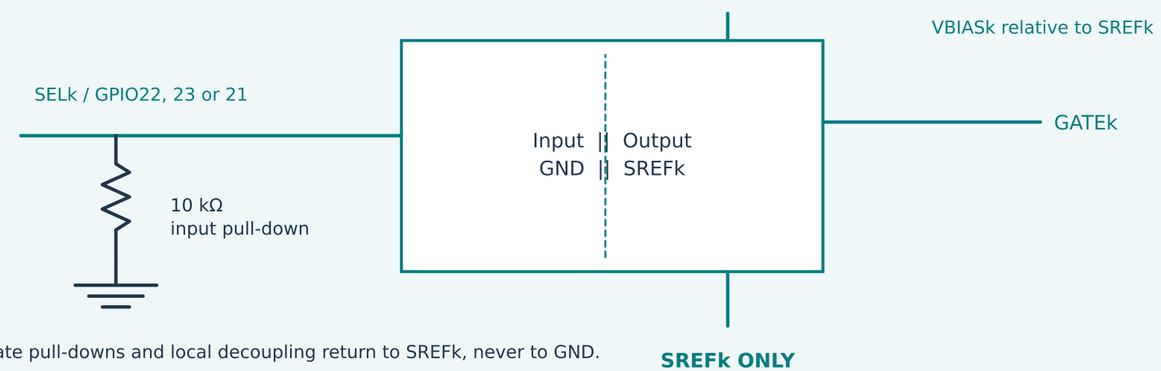
ACSELk: two opposing N-MOSFETs, common sources



GATEk — one small gate resistor per FET

Each body diode points from the joined sources toward its outer drain.

Uk: isolated gate driver + floating DC bias



Gate pull-downs and local decoupling return to SREFk, never to GND.

Average current + separate fault path

K+ / K− are Kelvin connections across RS.

ADC: sense amplifier → low-pass → ADC1
3.3 V supply; ground / VZERO on logic GND.
Scale and limit output to the configured ADC range.

FAULT: fast differential sense → comparator
→ latch → FAULT_OK; latch starts in INHIBIT.
ARM is LOW until one selector is ready; GPIO TBD.

Connection and validation notes

1. SREF0, SREF1 and SREF2 are separate floating nodes.
   Each selector needs its own DC-capable bias/driver.

2. U1 output return and local bypass capacitors go to RET.
   Its logic side stays on GND; isolated bias avoids a bypass.

3. RS measures combined return / input-inductor current.
   It does not measure piezo RMS or QMAIN peak separately.

4. Keep the chosen selector ON during bursts. Change it
   only after drive stops and measured ring-down ends.